Arista Access Node (HEB-ACC-11) Optical Transport Failure (ALARM_4415) - Major

Node: HEB-ACC-11 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_4415

Description

The Optical Transport Failure on the Arista Access Node (HEB-ACC-11) at the Hebbal Edge site is primarily attributed to the degradation of SFP/XFP transceivers or fiber cuts leading to a reduction in Rx/Tx power. This failure domain is more pronounced in the Access role due to the reliance on SFP/XFP transceivers for data transmission between the node and the Optical Transport Network (OTN). The Arista EOS (Extensible Operating System) utilizes a proprietary protocol for SFP/XFP management, which can lead to misconfiguration or hardware malfunctions. In addition, environmental factors such as temperature fluctuations, humidity, or physical stress on the fiber cables can also contribute to the failure. Furthermore, the lack of proper Quality of Service (QoS) settings or inadequate SFP/XFP insertion/extraction procedures can further exacerbate the issue.

Pre-requisite Checks

- Dispatch remote hands to the Hebbal Edge site to verify the physical integrity of the fiber cables and SFP/XFP transceivers.

- Check the status of the redundant link to ensure that it is functioning correctly.

- Verify that the Arista Access Node (HEB-ACC-11) is running the latest software image and has the most recent SFP/XFP database.

- Ensure that the OTN is functioning correctly and has adequate power and cooling.

Potential Root Causes

Cause	Probability
SFP/XFP Transceiver Degradation	High
Fiber Cuts	Medium
Rx/Tx Power Loss	Low
Misconfigured SFP/XFP Management Protocol	Medium
Environmental Factors (Temperature, Humidity, Physical Stress)	Low
Inadequate QoS Settings	Low
Improper SFP/XFP Insertion/Extraction Procedures	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface tengigabitethernet 1/1/1 detail, Expected Output: Interface tengigabitethernet 1/1/1, Hardware: \"10G Ethernet\", Address: 00:11:22:33:44:55 (bia 00:11:22:33:44:55), Description: This command isolates the interface and its associated hardware details, including the address and bia (burned-in address).

Command: show sfp status, Expected Output: SFP Status: Operational, Transceiver Type: 10G-SR, Description: This command provides information about the SFP/XFP transceiver, including its status and type.

Command: show environment, Expected Output: Temperature: 25.0°C, Humidity: 60%, Description: This command displays environmental information, including temperature and humidity levels.
```

Resolution Steps

Step 1: Step: Verify SFP/XFP Transceiver Status, Command: show sfp status, Description: Verify the status of the SFP/XFP transceiver and take corrective action if it is not operational.

Step 2: Step: Replace SFP/XFP Transceiver, Command: insert sfp/xfp 0/1/1 transceiver-type 10G-SR, Description: Insert a new SFP/XFP transceiver and configure it for the correct type and speed.

Step 3: Step: Verify Interface Status, Command: show interface tengigabitethernet 1/1/1 detail, Description: Verify that the interface is operational and has the correct address and bia.

Step 4: Step: Verify OTN Status, Command: show otu status, Description: Verify that the OTN is functioning correctly and has adequate power and cooling.

Step 5: Step: Verify QoS Settings, Command: show qos policy, Description: Verify that the QoS settings are adequate for the traffic flowing through the node.

Post-Check Verification

```
Command: show interface tengigabitethernet 1/1/1 detail, Expected Output: Interface tengigabitethernet 1/1/1, Hardware: \"10G Ethernet\", Address: 00:11:22:33:44:55 (bia 00:11:22:33:44:55), Description: Verify that the interface is operational and has the correct address and bia.

Command: show sfp status, Expected Output: SFP Status: Operational, Transceiver Type: 10G-SR, Description: Verify that the SFP/XFP transceiver is operational and has the correct type and speed.

Command: show otu status, Expected Output: OTU Status: Operational, Power: 1.2W, Cooling: OK, Description: Verify that the OTN is functioning correctly and has adequate power and cooling.
```

Roll-back Procedures

- Step: Remove New SFP/XFP Transceiver, Command: remove sfp/xfp 0/1/1, Description: Remove the new SFP/XFP transceiver and return to the original configuration.

- Step: Restore Original SFP/XFP Transceiver, Command: insert sfp/xfp 0/1/1 transceiver-type original, Description: Restore the original SFP/XFP transceiver and configure it for the correct type and speed.

- Step: Restore Original QoS Settings, Command: set qos policy original, Description: Restore the original QoS settings and verify that they are adequate for the traffic flowing through the node.